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A Sensorless Control of PMSM Using Hybrid Fuzzy–Logic Adaptive Sliding Mode Approach*

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SUMMARY A sensorless control method based on a super-twisting sliding mode observer (STSMO) and a second-order generalized integrator phase-locked loop (SOGI-PLL) with fuzzy control is proposed. The fuzzy control algorithm is integrated into the SOGI-PLL to reduce chattering and improve the accuracy of the rotor position estimation of the permanent magnet synchronous motor (PMSM). Firstly, the basic principles and mathematical models of the STSMO are explained in detail. Secondly, the basic concepts of the SOGI-FLL and the PLL are introduced, along with their applications in sensorless control. Then, based on the proposed method, a PMSM control strategy is developed following system modeling, observer design, and control strategy optimization. The effectiveness of the proposed method is verified through a comparative analysis between the optimized control strategy and the initial control strategy. The experimental results demonstrate that the proposed STSMO-SOGI-PLL method reduces velocity fluctuations to 40r/min and effectively eliminates steady-state estimation errors.

key words: Permanent magnet synchronous motor, Super twisting sliding mode, Second-order generalized integrator, Fuzzy control.

1. Introduction

In modern industrial applications, the demand for motor control accuracy and system reliability continues to grow steadily. The PMSM has become the preferred actuator in many fields due to its simple structure, high efficiency, and fast dynamic response. However, high-precision sensors are expensive, highly susceptible to environmental factors, and difficult to install. Sensors with low accuracy exhibit issues such as slow response speed and significant acquisition errors in practical engineering applications. This not only increases the system's complexity and cost but also restricts its applicability in extreme environments. To address these issues, sensorless control technology is emerging as a research hotspot in the field of motor control[1]–[4]. Sensorless control technology leverages the motor's physical characteristics and electrical parameters to accurately estimate rotor position and velocity, thereby eliminating the reliance on position sensors[5]. Researchers have developed various sensorless control algorithms, including those based on back electromotive force[6], extended Kalman filter[7], [8], model reference adaptive systems[9]–[11], adaptive algorithms[12] and variable structure techniques[13].

The sliding mode observer (SMO) [14]–[18] is widely recognized as a robust observer due to its insensitivity to system structure and parameter uncertainties[19], as highlighted in [20]. However, the SMO can generate high-frequency

noise during the design of the sliding surface, which may degrade the performance of the observer. In [21], adaptive algorithms are combined to design a novel adaptive sliding mode observer for estimating the rotor position and speed of a motor, and the convergence of the algorithm is proved using the Lyapunov theory.

The SMO is a nonlinear observer. Although traditional sliding mode control is characterized by strong robustness and good dynamic performance, it is affected by the chattering phenomenon, which is a challenging issue requiring resolution in sliding mode control. However, the Super Twisting algorithm is proposed in this article as an efficient second-order sliding mode control strategy, and it is used to estimate the back electromotive force of the PMSM due to its fast convergence speed and low chattering phenomenon[22]–[24]. The observer achieves a stable convergence of sliding mode variables and their first-order derivatives to 0. The sign function is embedded in the integral term, and the chattering phenomenon is effectively suppressed. The stability of the STSMO is analyzed, and the parameter range for algorithm convergence is determined, demonstrating significant potential in the field of sensorless control[25].

To further improve the observation accuracy and dynamic response of the system, a SOGI combined with a frequency lock loop (FLL) is introduced in this paper [26], [27]. The SOGI is often applied to the extraction of synchronous signals in power grids. The fundamental frequency signal is extracted when the SOGI resonance frequency matches the input signal frequency[28]. A more stable and accurate control signal is provided for the system. Compared with traditional low-pass filters, significant advantages in suppressing high-frequency noise and ripple are demonstrated by the SOGI–PLL, and clearer, more reliable signals are provided for sensorless control systems. Among them, a second-order fuzzy control algorithm is added to the SOGI and the fuzzy controller. we designed can adaptively adjust the gain k based on the real-time performance indicators of the system, such as error e and error change rate ec . The system is enabled by this adaptive mechanism to better cope with rapid changes and various disturbances.

A super twisting algorithm is proposed in this article for the sensorless control of the PMSM, which is combined with a sliding mode control strategy and integrated with the SOGI-PLL technology. An improved sliding mode observer method is developed using a fuzzy control algorithm. High-precision estimation of the PMSM rotor position and velocity is achieved with this strategy, which also exhibits

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strong anti-interference performance and system robustness, while the chattering problem in traditional sliding control systems is effectively mitigated. Through in-depth theoretical analysis and simulation experiments, sliding vibration in the PMSM is reduced, and rotor position estimation accuracy is improved.

2. Design of the Super Twisting Sliding Mode Observer

2.1 The mathematical model of the PMSM

A static reference coordinate system (α - β coordinate system) is used in this article to model permanent magnet synchronous motors. The specific expression is provided as follows[29].

$$\begin{cases} \frac{d\hat{i}_\alpha}{dt} = -\frac{R}{L_s}\hat{i}_\alpha + \frac{1}{L_s}u_\alpha - \frac{1}{L_s}e_\alpha \\ \frac{d\hat{i}_\beta}{dt} = -\frac{R}{L_s}\hat{i}_\beta + \frac{1}{L_s}u_\beta - \frac{1}{L_s}e_\beta \end{cases} \quad (1)$$

where, u_α and u_β are the stator voltage components of α -axis and β -axis, and i_α and i_β are the stator current components of the α -axis and β -axis, R is the stator resistance and L_s is the inductance. e_α and e_β is the component of the back electromotive force of the α -axis and β -axis.

$$\begin{cases} e_\alpha = -\psi_f \omega \sin \theta \\ e_\beta = -\psi_f \omega \cos \theta \end{cases} \quad (2)$$

where, ψ_f is the back electromotive force constant, ω is the rotor electrical angular velocity, and θ is the rotor electrical angular position.

2.2 Super Twisting Algorithm

$$\begin{cases} \dot{x}_1 = -k_1|x_1|^{\frac{1}{2}} \tanh(\bar{x}_1) + x_2 + \rho_1(x, t), \\ \dot{x}_2 = -k_2 \tanh(\bar{x}_1) + \rho_2(x, t), \end{cases} \quad (3)$$

where, x_1 and x_2 are the state variables of the system, and $\bar{x}_i$ is the error between the estimated state value and the actual value. ρ_i is the perturbation term of the i -state variable. k_i is the sliding mode coefficient of a state variable. The function $\tanh()$ is adopted as a continuous approximation of the sign function, which provides smoothness while retaining similar switching behavior.

In order to meet feasible system dynamics and stability, the sliding mode coefficients in (3) need to satisfy:

$$k_1 > 2\delta_1, \quad k_2 > k_1 \frac{5\delta_1 k_1 + \delta_1^2}{2(k_1 - 2\delta_1)} \quad (4)$$

where, δ_1 is any normal number, and the perturbation term in (3) must satisfy.

$$|\rho_1| \leq \delta_1 |x_1|^{1/2}, \quad \rho_2 = 0 \quad (5)$$

In sliding mode control, the selection of sliding mode

coefficients is crucial. It needs to be set sufficiently large to ensure that the system state can quickly reach the sliding mode surface within a predetermined time and remain stable on that surface. Helps to improve the convergence rate of the system under sliding mode control. Usually, the principle of selecting gain is to make it sufficiently large so that the system state can quickly approach the sliding surface and maintain a stable state after reaching that surface.

2.3 Super Twisting Sliding Mode Observer

The form of the sensorless control algorithm for the permanent magnet synchronous motor based on the traditional Super Twisting sliding mode observer is presented in (6).

$$\begin{cases} \frac{d\hat{i}_\alpha}{dt} = -\frac{R}{L_s}\hat{i}_\alpha + \frac{1}{L_s}u_\alpha + \frac{1}{L_s} (k_1|\hat{i}_\alpha|^{1/2} \tanh(\hat{i}_\alpha) + \int k_2 \tanh(\hat{i}_\alpha) dt) \\ \frac{d\hat{i}_\beta}{dt} = -\frac{R}{L_s}\hat{i}_\beta + \frac{1}{L_s}u_\beta + \frac{1}{L_s} (k_1|\hat{i}_\beta|^{1/2} \tanh(\hat{i}_\beta) + \int k_2 \tanh(\hat{i}_\beta) dt) \end{cases} \quad (6)$$

where, $\hat{i}$ represents the estimated current, $\bar{i}$ represents the error currently, k_1 and k_2 are the sliding coefficients.

The state equation of stator current estimation error can be obtained by subtracting equation(1) from equation(6):

$$\begin{cases} \frac{d\bar{i}_\alpha}{dt} = -\frac{R}{L_s}\bar{i}_\alpha + \frac{1}{L_s}\bar{e}_\alpha + \frac{1}{L_s} (k_1|\bar{i}_\alpha|^{1/2} \tanh(\bar{i}_\alpha) + \int k_2 \tanh(\bar{i}_\alpha) dt) \\ \frac{d\bar{i}_\beta}{dt} = -\frac{R}{L_s}\bar{i}_\beta + \frac{1}{L_s}\bar{e}_\beta + \frac{1}{L_s} (k_1|\bar{i}_\beta|^{1/2} \tanh(\bar{i}_\beta) + \int k_2 \tanh(\bar{i}_\beta) dt) \end{cases} \quad (7)$$

When the estimation error reaches the sliding mode, the sensorless control system reaches stability, and the estimated current value is approximately equal to the actual current value. At this time, the estimated back electromotive force obtained by equivalent control is as follows:

$$\begin{cases} \hat{e}_\alpha = -k_1|\hat{i}_\alpha|^{1/2} \tanh(\hat{i}_\alpha) + \int k_2 \tanh(\hat{i}_\alpha) dt \\ \hat{e}_\beta = -k_1|\hat{i}_\beta|^{1/2} \tanh(\hat{i}_\beta) + \int k_2 \tanh(\hat{i}_\beta) dt \end{cases} \quad (8)$$

After estimating the back electromotive force, the rotor position angle and rotor speed can be calculated using equations (9) and (10) respectively:

$$\hat{\theta} = -\arctan\left(\frac{\hat{e}_\alpha}{\hat{e}_\beta}\right) \quad (9)$$

$$\hat{\omega} = \frac{d\hat{\theta}}{dt} \quad (10)$$

To ensure the reliability of the above observer and the accuracy of the estimated states, it is necessary to analyze its stability and convergence properties. The stability of the proposed adaptive sliding mode observer is examined via Lyapunov theory. Let the estimation errors be $\tilde{i}_\alpha = i_\alpha - \hat{i}_\alpha$ and $\tilde{i}_\beta = i_\beta - \hat{i}_\beta$. Consider the Lyapunov candidate:

$$V = \frac{1}{2}\tilde{i}_\alpha^2 + \frac{1}{2}\tilde{i}_\beta^2 \quad (11)$$

which is positive definite for nonzero estimation errors.

Differentiating V along the error dynamics and applying the super-twisting algorithm yields:

$$\dot{V} \leq -k_\alpha |\tilde{i}_\alpha| - k_\beta |\tilde{i}_\beta| < 0 \quad (12)$$

for $k_\alpha, k_\beta > 0$, ensuring finite-time convergence of the estimation errors to zero.

Furthermore, the introduction of SOGI-PLL and fuzzy gain adaptation improves robustness against parameter uncertainties and external disturbances, enabling stable real-time performance.

3. Design of the SOGI-FLL

Super Twisting sliding mode control will use a low-pass filter to filter the back electromotive force and eliminate resonance caused by high-frequency switches, which will cause phase lag and affect the stability of the system. This article uses the SOGI-FLL to construct a filter for phase synchronization in a locked frequency loop. Figure 1 shows the structure of the SOGI-FLL filter.

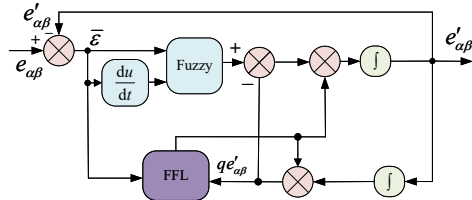


Fig. 1 The SOGI-FLL Structure Diagram.

From Figure 1, it can be concluded that $e'_{\alpha\beta}$ and $qe'_{\alpha\beta}$ are orthogonal to each other and are related to the corresponding SOGI closed-loop transfer function $D(s)$, $Q(s)$

$$\begin{cases} D(s) = \frac{e'_{\alpha\beta}}{e_{\alpha\beta}} = \frac{k\omega_f s}{s^2 + k\omega_f s + \omega_f^2} \\ Q(s) = \frac{qe'_{\alpha\beta}}{e_{\alpha\beta}} = \frac{k\omega_f^2}{s^2 + k\omega_f s + \omega_f^2} \end{cases} \quad (13)$$

where, ω_f is the resonant frequency of the filter. $D(s)$ can be regarded as a bandpass filter, whose bandwidth is determined only by the gain k and is independent of the resonant frequency. The center frequency of the bandwidth is the resonant frequency of SOGL. The smaller the gain k , the smaller the bandwidth, and the better the frequency selectivity, but the longer the steady-state time. $Q(s)$ can be regarded as a low-pass filter, where the output signal $qe'_{\alpha\beta}$ always lags behind $e'_{\alpha\beta}$ and outputs 90° .

Given the above filter characteristics, the convergence behavior of the proposed SOGI-based observer can be further evaluated via its error transfer function, relating the input estimation error $E_{in}(s)$ to the output estimation error $E(s)$: $E_{in}(s)$ and the output estimation error $E(s)$ is:

$$E(s) = \frac{s^2 - \omega_0^2}{s^2 + k\omega_0 s + \omega_0^2} E_{in}(s) \quad (14)$$

where $\omega_0 > 0$ is the resonant frequency and $k > 0$ is the gain

constant. Applying the final-value theorem gives:

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} \frac{s(s^2 - \omega_0^2)}{s^2 + k\omega_0 s + \omega_0^2} E_{in}(s) \quad (15)$$

If $E_{in}(s)$ is bounded and all poles of $sE(s)$ lie strictly in the left-half plane, then $\lim_{t \rightarrow \infty} e(t) = 0$, ensuring asymptotic convergence for $k > 0$ and $\omega_0 > 0$.

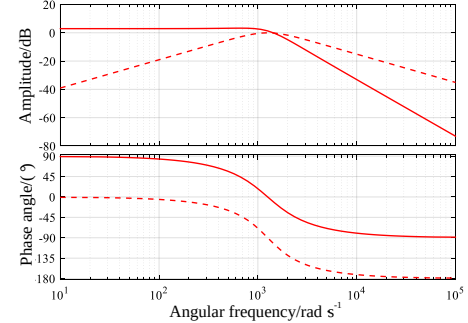


Fig. 2 Amplitude-Frequency Characteristic Curve.

As shown in Fig. 2, the amplitude-frequency characteristic of the SOGI filter illustrates the effect of the gain k on the filter's bandwidth and frequency selectivity. Under the product of the estimation error $\tilde{\varepsilon}$ in the SOGI and the orthogonal output $qe'_{\alpha\beta}$, ε_f is used as the input of the FLL module. When the frequency of the sine input signal $e_{\alpha\beta}$ is less than the FLL output frequency ($\omega < \omega_f$), $qe'_{\alpha\beta}$ and $\tilde{\varepsilon}$ are in phase, $\varepsilon_f > 0$, when the frequency of the input signal is greater than the FLL output frequency ($\omega > \omega_f$), $qe'_{\alpha\beta}$ and $\tilde{\varepsilon}$ are in opposite phases, $\varepsilon_f < 0$, when $\omega = \omega_f$, $\varepsilon_f = 0$. The integrator with negative gain $-\tau$ in FLL gradually adjusts the output frequency ω_f according to the change, ultimately making $\omega = \omega_f$.

In the SOGI-PLL system, a fuzzy control algorithm is adopted to dynamically adjust the gain k for improved transient response and enhanced robustness against parameter variations and external disturbances. The fuzzy controller is a second-order regulator with two input variables: the system output error $e = r - y$ and the error change rate $e_c = e(k) - e(k-1)$, where r is the reference signal and y is the system output. These inputs jointly describe the current deviation and the dynamic trend of the system response.

Triangular membership functions are selected for their simplicity, computational efficiency, and suitability for human-like reasoning. Each input domain is divided into five fuzzy sets: negative large (NB), negative small (NS), zero (ZO), positive small (PS), and positive large (PB), with $\mu(x) \in [0, 1]$ denoting the membership degree.

Based on these fuzzy sets, the control logic is expressed as IF-THEN rules. The fuzzy inference process computes the output membership $\mu(u)$ via the min-max method, and the crisp control signal is obtained by centroid defuzzification:

$$u^* = \frac{\int u \cdot \mu(u) du}{\int \mu(u) du} \quad (16)$$

In the proposed controller, the inputs e and e_c are

mapped to the output variable u , representing the adaptive gain adjustment factor of the observer. Fig. 3 illustrates the membership function of them. Fig. 3 illustrates the membership function of e . To ensure consistency with e_c , the horizontal axis for e in Fig. 3 has been scaled down by a factor of 20.

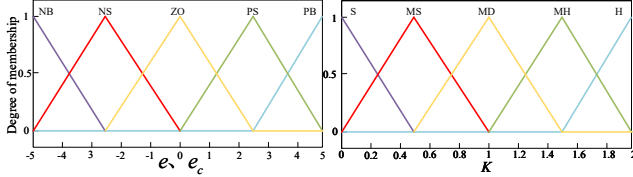


Fig. 3 The membership function of input and output.

According to the input–output relationship and STSMO–SOGI–PLL control objectives, 25 fuzzy inference rules were designed, as shown in Table 1.

Table 1 Fuzzy control rule table.

e/e_c	NB	NS	ZO	PS	PB
NB	H	MH	MD	MS	S
NS	MH	MD	MS	S	MS
ZO	MD	MS	S	MS	MD
PS	MS	S	MS	MD	MH
PB	S	MS	MD	MH	H

Through fuzzy control, the gain k is adaptively tuned in the SOGI–PLL system to optimize the dynamic response, improve robustness against parameter variations and external disturbances, and enhance the filtering effect as well as overall system performance.

4. Position Estimation of Phase-locked loop Rotor

Through phase-locked loop technology, phase synchronization between two frequency signals is achieved. The phase difference between the input signal and the output signal is eliminated by comparing their phases and adjusting the output frequency until the desired frequency is reached. During this process, the phase difference of the signal is maintained at a constant value, indicating that the phase has been successfully locked. Furthermore, the phase-tracking characteristics of phase-locked loops are utilized to estimate the position and velocity of motor rotors, significantly reducing errors caused by disturbances and improving measurement accuracy. The phase-locked loop control block diagram applied to the PMSM is presented in Figure 4.

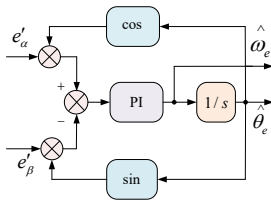


Fig. 4 Implementation Block Diagram of PLL.

In the sensorless system of a PMSM, the phase-locked loop is employed to estimate the electrical angle, which is used as the electrical angle for coordinate transformation. The speed estimated by the phase-locked loop is divided by the number of pole pairs of the motor to calculate the actual speed of the motor.

The phase-locked loop technology is employed to accurately estimate the rotor position. However, the estimated angular frequency contains significant noise, which degrades the dynamic performance of speed observation and necessitates filtering.

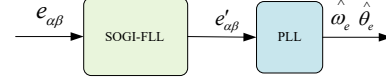


Fig. 5 Schematic diagram of SOGI-FLL-PLL.

5. Experimental Validation

A sensorless vector control system for the PMSM is built based on the super twisting sliding mode observer and SOGI–PLL, as shown in Figure 6. To verify the effectiveness of the proposed sensorless method, experimental tests are conducted on the motor platform, as shown in Figure 7.

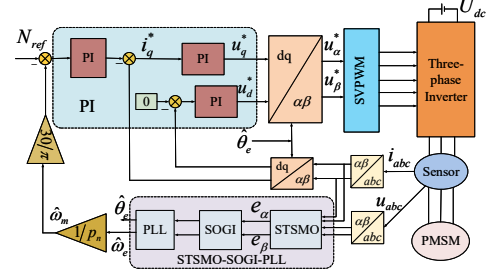


Fig. 6 Structure diagram of the PMSM sensorless control system.

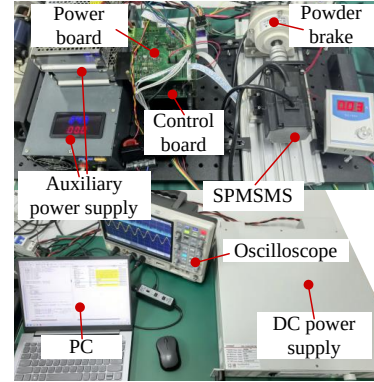


Fig. 7 Schematic diagram of the experimental platform.

Table 2 Parameters of the prototype and inverter.

Parameters	Value
Stator resistance R (Ω)	1.6
Inductance L_s (H)	0.005075
Pole Pairs	4
Permanent magnet flux λ_f (Wb)	0.0825
Rated speed n_N (r/min)	3000
Rated power P (kW)	0.2
Rated voltage U (V)	220
Rated current I (A)	2.1
Rated torque T_e (N·m)	0.64

The parameters of the PMSM used in the system are defined as Table 2, and vector control is implemented using the $i_d^* = 0$ control mode. The PI controller gains are $K_p = 5$ and $K_i = 1750$, and the PLL gains are $K_p = 30$ and $K_i = 300$, which are kept identical for all compared methods.

In Fig. 8, three observers (SMO, STSMO, and STSMO–SOGI–PLL) are compared in terms of observed speed, speed

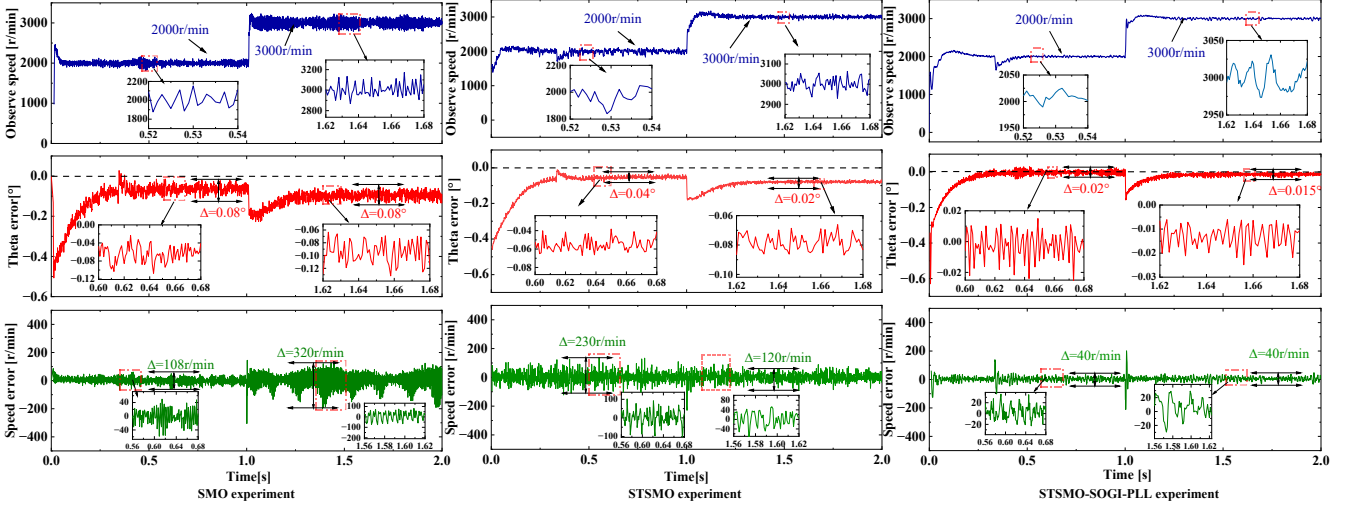


Fig. 8 Experimental data graphs of three methods

error, and angle (θ) error to assess disturbance rejection and steady-state stability. The SMO shows the poorest stability, with a steady-state θ -error of $\Delta \approx 0.08^\circ$, zero-crossing errors of 0.06° at 2000 r/min and 0.10° at 3000 r/min, and speed-error bands of $\Delta \approx 108$ r/min and $\Delta \approx 320$ r/min, respectively. The STSMO improves both accuracy and smoothness, reducing the θ -error to $\Delta \approx 0.04^\circ$ and $\Delta \approx 0.02^\circ$ and the speed-error to $\Delta \approx 230$ r/min and $\Delta \approx 120$ r/min at 2000 and 3000 r/min. The proposed STSMO-SOGI-PLL delivers the best performance, with θ -errors of $\Delta \approx 0.02^\circ$ and $\Delta \approx 0.015^\circ$, a constant speed-error band of $\Delta \approx 40$ r/min, and the smoothest response. At rated speeds of 2000 r/min and 3000 r/min, the theta errors at 0° are 0.06° and 0.1° for the SMO, 0.05° and 0.08° for the STSMO, and only 0.002° and 0.01° for the proposed STSMO-SOGI-PLL, respectively. When the reference speed steps from 2000 r/min to 3000 r/min at $t = 1$ s, it achieves a rise time of 0.006 s, overshoot of 3.1%, and steady-state error of 0.0002%, outperforming SMO (0.005 s, 8.0%, 0.04%) and STSMO (0.023 s, 5.9%, 0.02%).

To further evaluate estimation accuracy, the Root Mean Square Error (RMSE) of speed and position estimation is calculated under a step-speed reference from 2000 to 3000 r/min. In the 2000 r/min region (0–1 s), the speed RMSEs are 102.59, 52.91, and 35.42 r/min for SMO, STSMO, and STSMO-SOGI-PLL, respectively; in the 3000 r/min region (1–2 s), they are 54.91, 28.92, and 20.44 r/min. For position estimation, the angle RMSEs are 0.1377° , 0.1281° , and 0.1035° in 0–1 s, and 0.1141° , 0.0956° , and 0.0313° in 1–2 s. The significant RMSE reduction and enhanced transient response achieved by the STSMO-SOGI-PLL, especially at high speed, confirm its superior phase-tracking accuracy, fast dynamics, strong decoupling ability, and robustness to parameter variations and load disturbances.

As shown in Fig.9, STSMO reduces THD from 466.08% to 758.64%; adding SOGI-PLL further lowers it to 22.72%, which is 735.92% lower than SMO and 443.36% lower than STSMO.

6. Conclusion

An improved sliding mode observer based on STSMO-

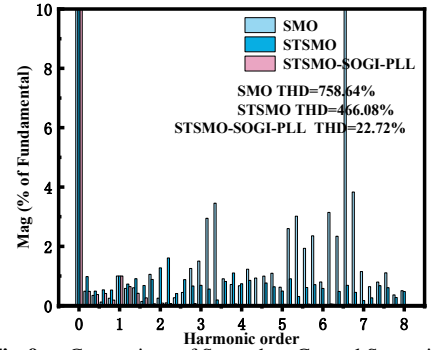


Fig. 9 Comparison of Sensorless Control Strategies.

SOGI-PLL is designed in this study, and the following key innovations are introduced.

1. Integration of STSMO, SOGI-PLL, and Fuzzy Control to reduce high-frequency noise and sliding mode chattering.
2. The observer achieves high-precision rotor position estimation for PMSM through an optimized algorithm.
3. The method significantly improves steady-state accuracy and dynamic response.

Experimental and simulation results verify the superior performance of the STSMO-SOGI-PLL algorithm, demonstrating effective chattering suppression, enhanced estimation accuracy, and robustness. The proposed model outperforms traditional methods in rotor position accuracy, steady-state performance, and dynamic response.

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